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Aidan Hong

Predicting Protein Structures

Richard Wang

The Marvels of Combat Robotics

Arthur Liang

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Elephant Shrews

Mary Liang

Of the 19 different types of elephant shrews, the Golden Rumped elephant shrew is alone at the edge of being endangered due to habitat fragmentation. Also known by the name of Sengis, they are identified with their long, sensitive snouts with characteristics based on their species. Their slim bodies make them agile and they run up to 18 mph with their long hind legs. These curious creatures are diurnal (active during the day), and when alarmed, their ears and eyes grow large to take in their environment for danger.

The elephant shrew's typical diet is made up of small insects, like ants and termites by constantly turning over leaves and soil. They live in the southern parts of Africa in Namibia, Botswana, and South Africa. As most people might guess, elephant shrews are shrews, but recent evidence points to the fact that they are closer to certain groups of African mammals like elephants, sea cows, and armadillos. Though their appearance may not be as similar, elephants are taken into this example. Elephant shrews have an extraordinary sense of smell and hearing, like elephants who can hear as low as 14 Hz (human's lowest: 20hz) and up to 12,000 Hz elephants use their trunks as a useful tool to have around due to their size, which makes them hard to hunt by several animals.

Regarding an elephant shrew's sleeping quarters, which some are situated as nests on forest floors, it lives in burrows, dirt cavities, rock nooks, termite mound slits, or under tree logs. They feed animals, including snakes and preparatory birds, like vultures and eagles. Elephant shrews typically live between 1 to 2 years in the wild, but their whole lifespan can be 3 to 4 years under good shelter.

Antimicrobial Resistance and its Threat to Public Health

Yidian Wang

Antibiotics are medicines that kill and slow bacterial growth to fight infections and disease, becoming an important foundation of our health. But what happens when the bacteria fight back and learn to resist these medications? This article will explore the misuse of agriculture and medicine, how it relates to bacterial evolution, and how it has become a problem for our health worldwide.

Antimicrobial resistance, also called AMR, occurs when bacteria, viruses, fungi, and protozoa evolve to counteract the antibodies that are designed to kill them. This happens due to several reasons, including the over-prescription of antibiotics and misuse in agriculture, causing spontaneous mutation of unwanted bacteria. When we use antibiotics, some resistant bacteria survive and multiply, resulting in more of these resistant bacteria. Constant use of antibiotics increases the spread of these bacteria, which leads to the need for more expensive or risky alternatives for treatment. The World Health Organization (WHO) recognizes AMR as a global health threat and is trying to combat it through research and surveillance. Although the effects are severe, resulting in nearly 5 million deaths a year, preventative measures exist to slow the spread of this dangerous bacteria.

Strategies such as using narrow-spectrum antibiotics, improving hygiene habits, and vaccinations are all effective preventative measures to avoid the harmful spreading of AMR bacteria. Moreover, the impacts of AMR can still be avoided by the responsible use of prescriptions by healthcare providers and more public awareness and education. Currently, there is ongoing research to create effective treatments for resistant bacteria and find other ways to fight infections without relying solely on antibiotics

Revolutionizing Technology: Predicting Protein Structures

Richard Wang

The use of artificial intelligence has enabled humans to do many things better than before. Recently, this has been best exemplified by the AI system, AlphaFold, which earned a Nobel prize for chemistry in 2024. Developed by David Baker, Demis Hassabis, and John M. Jumper, this algorithm is able to predict structures for all known proteins. Hassabis and Jumper, the main architects of this revolutionary algorithm, are able to predict protein structures from only the sequence of linked amino acids.

Proteins, which are composed of amino acids, have a hierarchy of structures and fold into different specific shapes. The function of a protein is determined by its structure. With this ability, scientists can understand diseases and how they are caused by dysfunctional proteins. It also helps with drug discovery and being able to target specific proteins involved in diseases and cancer.

Preceding this discovery, protein structures were determined by methods prone to human error including electron microscopy and x-ray crystallography. In addition, for years, methods such as molecular dynamic simulations were not effective due to the complexity of predicting protein structure. However, AlphaFold is able to find a correlation between amino acid sequence and protein structure to predict an unknown structure. AlphaFold first debuted 6 years ago, and significant changes were made to improve the range of protein sequences it can predict. Now, it is capable of predicting a whopping number of 200 million protein sequences from all life domains. Since AlphaFold is an AI research tool, it needs support from human experts. With the combination of human judgment and AlphaFold, there is a lot to learn about protein structure!

The Psychology of Personality

Denise Lee

The question of what makes a person introverted or extroverted has been fascinated by psychologists who have developed several theories to explain the psychology behind personality. The four psychology perspectives that go in-depth on the psychology of personality are psychoanalysis, humanistic, trait, and social cognitive perspectives. Each perspective offers unique insights on why people are the way they are.

The psychoanalytic perspective, developed by Sigmund Freud, focuses on the unconscious reason why an individual behaves the way they do. It focuses on past experiences, unconscious thoughts, and internal conflicts. According to the psychoanalytic perspective, introverted individuals may be introverted because, from childhood, they may have dealt with social anxiety and kept their emotions and thoughts hidden. Additionally, introverted people may lack confidence because they weren't praised as much growing up. On the other hand, extroverted individuals may be extroverted because they grew up in an extremely supportive environment where being social was encouraged. They may be self-centered and confident because they were always praised growing up, allowing them to have fewer internal conflicts and express themselves more freely.

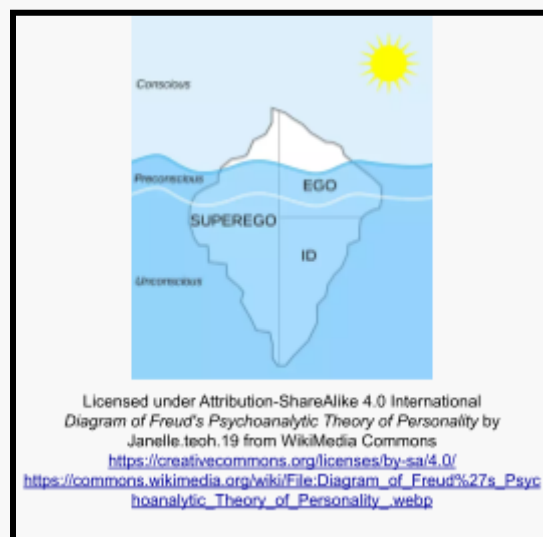
The humanistic perspective, originated by Abraham Maslow and Carl Rogers, focuses on the individual as a whole and it emphasizes free will and personal growth. Based on the humanistic perspective, people may be introverted because they choose to be and they do not enjoy the company of other people as much. Besides, they may simply prefer calm environments to noisy ones because the former provides an opportunity to think. In contrast, extroverted people may be extroverted due to their love

for talking and interaction. They may enjoy social stimuli, activities, and environments because that's what makes them happy!

The trait perspective, popularized by Ramond Cattell and Hans Eysench, emphasizes how behavior and personality are made up of traits. According to the trait theory, it can be argued that a person is introverted or extroverted based on the traits they possess. Introverted people tend to be socially reserved, quiet, calm, comfortable being alone, and observant, while extroverted people tend to be outgoing, talkative, energetic, and optimistic.

Lastly, the social cognitive perspective, founded by Albert Bandura, deals with social interactions and the environment that shapes behaviors. The cognitive theory of personality implies that introverted individuals are introverted because their environment neglects them or makes them feel uncomfortable. People may have overwhelmed them or made them feel unwanted, causing them to suppress their emotions and become extremely reserved. Extroverted people may be extroverted because the environment they surround themselves in is very supportive, giving them the confidence to socialize.

In conclusion, the origins of introversion and extroversion can be explained from various psychological perspectives, each offering a different viewpoint on the development of personality.



Sound Waves

Darren Wang

We often hear sounds from people or nature or the school bell or an iPhone notification. These sounds are processed through sound waves. Sound waves are waves of sound created either by us or other people. Sound is not only processed through waves by also vibrations that travel through materials like air, water, or solids.

Sound waves are longitudinal waves that result from the vibration of particles, such as gas, liquid, or solids. When an object vibrates, it creates compression and rarefaction, which allows energy from the vibrations to travel through the things so we can hear. The properties of sound waves such as frequency, amplitude, and speed, determine how we receive sound. Frequency, measured in hertz, determines the pitch of the sound we hear, while amplitude relates to the volume. In air, sound travels at approximately 343 meters per second, but this speed increases and becomes denser compared to water and solids.

The importance of sound waves extends and we use them in our everyday lives whether its communication or enjoyment of music. Sound wave technology can also be helpful for doctors. Sound wave technology uses high-frequency sound waves to paint a picture of the organs inside the body where the human eye can't see. Sonar technology also uses sound waves to detect objects underwater, to help in deep-sea marine research. These examples show how important sound waves are and how we use them as we move along.

As research continues to strengthen, new and innovative applications for sound waves are appearing. Scientists are exploring things such as “acoustic levitation”, which uses sound waves to lift small objects without physical contact, and focused ultrasound techniques for target drug delivery in medicine.

The Role of AI in Climate Change Solutions

Emily Ma

Artificial Intelligence has increasingly grown to be perceived as an essential tool in the fight against climate change. Because of the significant volumes of data being handled, AI will no doubt assist scientists and policymakers in deciding how environmental destruction can be contained. For example, AI-based models can monitor the levels of greenhouse gas emissions almost in real-time. Indeed, these allow governments and companies to monitor their performances and create more accountability for their commitments. Besides that, such systems can even predict future environmental conditions and, as such, allow leaders to initiate policies rather than just reactive ones proactively. This saves lives and resources by better planning in the most vulnerable regions. AI also goes a long way in optimizing renewable energy sources such as wind and solar power.

The AI system will monitor and illustrate the proper generation and storage of energy, from analyzing meteorological conditions to energy consumption data. In such a way, it maximizes efficiency by reducing waste, hence becoming more reliant on cost-effective renewable energy. Companies increasingly use AI due to the growing demand for renewable energy resources to make proper predictions about demand for renewable energy resources and proficiently manage the electricity grid. This becomes all the more important because renewable energy constitutes a big chunk of the world's energy mix. Besides energy management, AI made forecasting and the creation of responses against natural calamities such as hurricanes, floods, and wildfires due to climate change possible.

Soil Science

Audrey Wu

Soil Science is the study of soil as a natural resource, including its composition, formation, and functions of soil in various ecosystems. It merges aspects of biology, chemistry, physics, and geology to understand how soil supports life on Earth. Through soil science, people can gain insights into the soil's ability to sustain plant growth, regulate water, and interact with the atmosphere. Soil Science is critical for agriculture, environmental conservation, and even climate change.

The study of soil science begins with first understanding soil formation and classification. Soil forms through the weathering of rocks and organic matter decomposition, influenced by climate, organisms, topography, and time. This process results in diverse soil types, each with a unique physical and chemical property. Soil scientists classify soils into various orders and series to manage land use effectively and predict soil behavior.

Soil science also has a role in water regulation and filtration. Soils can store and purify water, influencing the hydrological cycle. The soil's ability to retain water affects groundwater recharge, flood prevention, and drought resilience. By studying soil hydrology, scientists can develop strategies to manage water resources efficiently and protect ecosystems from the effects of water-related issues such as flooding.

One of the essential functions of soil is its role in supporting plant growth. Soil provides plants with nutrients, water, and root anchorage. It acts as a reservoir of essential elements like nitrogen, phosphorus, and potassium, which are vital for plant health. Understanding soil properties, such as texture, structure, and pH, helps farmers optimize crop yields and develop sustainable farming practices that can allow us to better grow crops.

The Bifurcation Diagram: A Window into Chaos and Order

Eddie Zhang

The bifurcation diagram is a powerful visual tool that shows how a system's behavior changes as a parameter varies, transitioning from stability to chaos. It plays a key role in studying nonlinear systems, highlighting points where small changes cause shifts in long-term behavior. A well-known example is the logistic map, which models population growth through the equation $x_{n+1} = rx_n(1-x_n)$. As the growth rate r increases, the system moves from stable behavior to oscillations, and eventually to chaotic fluctuations, visualized through period-doubling bifurcations.

What makes the bifurcation diagram fascinating is its interplay between order and chaos. Even within chaotic regions, islands of stability appear, showing brief returns to periodic behavior. This reflects that chaos, though unpredictable, contains structure. Additionally, the Feigenbaum constant (around 4.669) governs the rate at which these bifurcations occur, a pattern seen across various systems in nature and science. The diagram also illustrates the sensitive dependence on initial conditions, a hallmark of chaos, where tiny differences in starting points lead to drastically different outcomes. This concept, known as the butterfly effect, limits long-term predictability. The insights from bifurcation theory have practical applications in fields like biology, neuroscience, economics, and climate science, helping to explain phenomena like population dynamics, brain activity patterns, and financial market volatility.

In addition to practical uses, the diagram raises philosophical questions about predictability. It shows that even deterministic systems can behave unpredictably, reminding us of the limits of control over complex systems.

The Forgotten Role of Fungi in Ecosystems

Brandon Pian

Fungi are often underestimated in ecosystems; however, their role is fundamental to maintaining a balance in an environment. Considering the concepts of degradation processes, their functions are not limited to those. Fungi are important in ecosystem functioning since they can interconnect diverse environmental components and serve as a way toward stability.

Fungi play a very important role, particularly in nutrient cycling. Decomposers like saprophytic fungi break down complex organic material and release into the soil crucial nutrients like carbon, nitrogen, and phosphorus. This is very important for soil fertility, supporting plant life, and ensuring ecosystem productivity.

Besides this, mycorrhizal fungi also develop mutualistic associations with plant roots. In return for receiving carbohydrates from plants, they aid in increasing the absorption of water and nutrients in plants, especially when nutrients are very low in the soil environment. Such mutualism is crucial to the health and resistance of plants.

Fungi also help regulate ecosystems concerning pest and pathogen control. They have been applied to a series of attempts toward bioremediation, related to the breakdown of environmental pollutants, which justifies their ecological importance.

In conclusion, fungi are vital in nutrient cycling, regulation of species, and robustness of ecosystems. The contribution that they make toward the working of ecosystems underlines their role in the health of the planet. An appropriate appreciation of the many important roles played by fungi helps deepen our appreciation for biodiversity itself and also informs conservation and management strategies.

The Marvels of Combat Robotics

Arthur Liang

If you have ever encountered the world of combat robotics, you'd know that it is a source of thrilling entertainment and destruction that satisfies something in the primitive side of our brains. Yet, behind the scenes, building combat robots is an extremely difficult endeavor, an intersection of advanced physics, programming, and engineering. Each discipline plays a crucial role in designing competitive and resilient fighting machines.

Physics is the foundation of combat robotics, determining the impact force, speed, and agility of the robot. Engineers consider many unique materials and their properties to balance strength, weight, and price (combat robots are NOT cheap). The principles of energy transfer, torque, and momentum are essential for designing effective weapon systems. They must also consider the center of gravity and the gyroscopic effects of spinning weapons to maintain stability during high-speed maneuvers and collisions, ensuring the robot can stay upright while delivering powerful attacks.

Programming is the brain behind the robot's movements and weapon systems. It involves writing code that allows the robot to respond to real-time inputs, whether controlled manually or autonomously. Combat robotics often employ advanced algorithms for navigation, target tracking, and weapon deployment. The code is optimized to handle quick decision making and split-second reactions to opponents' strategies.

Engineering blends all these aspects into a functioning machine. Mechanical engineers design robust chassis,



effective weapons, and efficient drive systems. Electrical engineers link together motors, batteries, and control systems, ensuring that the power demands of the robot's motors and weapons are met. Structural engineering choices, such as armor layout and weapon mounting, are made to withstand severe impacts while maximizing damage output. And, of course, builders must learn to repair and reconstruct their robot after a potentially devastating fight.

Overall, combat robotics is a multidisciplinary field where the perfect synergy of many subjects creates formidable machines capable of outmaneuvering and overpowering the competition. If you are interested, check out BattleBots or NHRL (National Havoc Robot League), as they have some of the best robot fighting tournaments out there.



Advances in Space Exploration

Kenny Wu

Since the start of human civilization, areas beyond the boundary of the sky have captivated our attention. The era of space exploration began in the mid-20th century, making prominent steps into the unknown with events such as the launch of Sputnik 1 in 1957, which advocated for the usage of artificial satellites, and the Apollo 11 mission in 1969, where humanity first imprinted their trace on the moon. These milestones toward outer space displayed the remarkable revolution of our technological capabilities and manifested our desire to explore the universe.

In recent years, the commercial spaceflight sector has played a significant role in space exploration, greatly altering our methods of accessing space. Key contributors include SpaceX, Blue Origin, and Virgin Galactic, each pioneering technologies that refine our ability to traverse outer space. SpaceX, in particular, has hit the headlines with its Falcon 9 rocket, featuring reusable technology that tremendously reduces the cost of launching payloads into orbit. The reusability trait of SpaceX rocket models allows trips to outer space to be economically viable and cost-efficient.

Furthermore, Mars has been the spotlight of human fascination and ambition for exploration. The planet is commonly believed to be humanity's next home in the solar system. In February 2021, NASA's Perseverance Rover successfully landed on the Martian surface, representing a significant leap forward in humanity's quest to understand the Red Planet. Equipped with advanced scientific instruments, Perseverance was tasked with searching for signs of microbial life and collecting soil and rock samples. All in all, humanity has made an enormous leap to unravel more mysteries of the universe, with the advancements made in recent years being both exhilarating and transformative. The rise of commercial spaceflight has opened up greater magnitudes of methods to outer space and improved the transparency of space exploration in general.

The Science of Music and the Brain: How Sound Shapes Our Minds

Natalie Dai

When we listen to music, sound waves trigger vibrations in the ear, which are converted into electrical signals that travel to the brain. This process is accountable for processing rhythm, melody, and harmony, and the auditory cortex and motor regions. In response to this process, dopamine, a chemical linked to pleasure and reward, is released by the brain. This explains why many people feel joy or an emotional connection when they listen to music they love. Moreover, familiar songs can evoke vivid memories by activating the hippocampus, a region linked to memory formation.

Music has an impact that goes beyond enjoyment and sentimentality. Studies have demonstrated that listening to music or playing an instrument improves cognitive abilities like memory, focus, and problem-solving. Because musicians have spent years coordinating complex motor and auditory activities, they have improved connectivity across the hemispheres of their brains. Improved executive function, a collection of abilities required for concentration, planning, and multitasking, has been connected to this increased brain connectivity.

Music therapy has demonstrated efficacy in clinical contexts in addition to its cognitive advantages. Music can help individuals with depression, Parkinson's disease, and Alzheimer's disease manage their symptoms and enhance their quality of life. Through the use of melody and rhythm, music therapy helps patients develop their motor skills by activating brain regions linked to mood management and movement.

The potential uses grow as researchers continue to examine the effects of music on the brain. The science of music demonstrates that sound is more than just an art form; it is a potent tool for enhancing mental and emotional well-being.

What is a MERN Stack?

Aidan Hong

Many websites offer a wide range of services, from allowing you to watch movies, buy food, and more. However, the technology behind such services is more complex than many expect. Some developers will utilize MongoDB, Express.JS, React.JS, and Node. JS(MERN) to create such an advanced website, taking care of both the front end and back end to create a full-stack application.

The first part of a MERN stack is MongoDB, which is used for back-end functions. MongoDB is a NoSQL database or a database that doesn't have any relations. A relation in a database is when data is linked to other data in another table. This means that data stored in a MongoDB database will not be linked to other data in that same database in any shape or form. Typically, most will use MongoDB to store usernames and password hashes for authentication. Password hashes are stored instead of passwords because password hashes are hard to break and typically need brute force tools like hashcat to crack. Most developers will typically use MongoDB, but if the developer needs a relational database and SQL, a popular alternative is PostgreSQL.

The second part of a MERN stack is Express.JS. Express is used to handle HTTP requests and responses, as well as manage data and implement logic for the server, where the back-end program is usually hosted. Express is used when the user requests data: express will handle which route will handle the data and respond with the appropriate data. Express is popular: chances are, if you click "inspect element" and look in the network tab, most data that you request will use express!

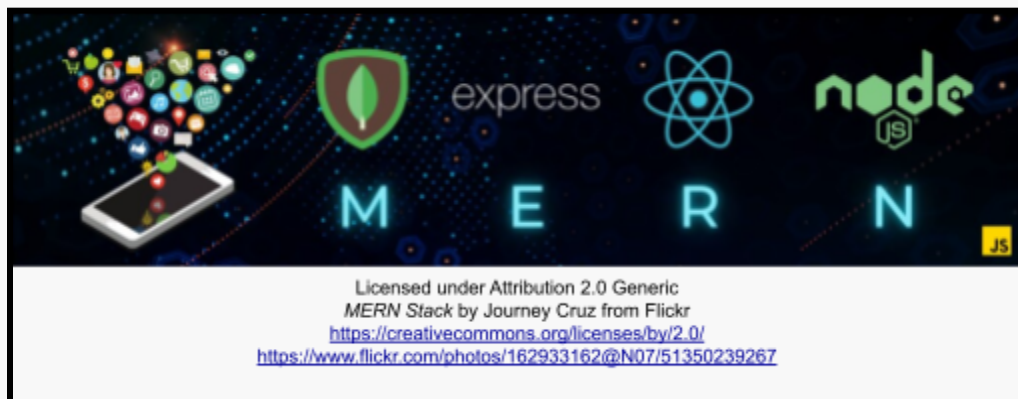
The third part of a MERN stack is React.JS. React is typically used for the front end. React offers a framework developers can use to create the front end of a webpage. The front end is typically used to show the content to the user, as well as show data from the backend in a presentable format, after all, it isn't fun to read raw JSON. React uses JSX

components, which include aspects of both JS and HTML. Their syntax is almost identical to HTML! While React.JS is great, there are also many other alternatives to React.JS, such as Next.JS, Vite, Angular, and more.

Finally, the last part of a MERN stack is Node.JS. Node.JS is used to create the overall backend. This is often used to run the actual code: while Express handles the logic and routes data appropriately, Node.JS will run the data and return the actual data.

Additionally, Node.JS is also used to call API's. API's are requests to other servers or websites, and can be used to expand the website's functionality without having to write much code: the only code needed is to call the API and format the data sent and received! There are also many other alternatives to Node.JS, such as Flask, Java, and Ruby on Rails.

All of these technologies combine to create a MERN Stack. MongoDB is used for authentication as well as to store data, while Express is used to handle logic and route data accordingly. React is used to create a great frontend while Node.JS is used to handle backend programming and works with Express to return data to the user. Combined, we get the powerful and stunning websites we all love and enjoy using.



Process of Photosynthesis

Owen Chen

All lifeforms need some source of energy to function within this world, with some receiving it by consuming others, while others simply create it themselves. One of the most common creations of this is through the process of photosynthesis; plants convert sunlight, water, and carbon dioxide into a usable form of power source for all organisms called glucose, and also release oxygen molecules into the air. Although we may not see it, the procedure of photosynthesis is the essential drive of the ecosystems that can be studied through microscopic lenses, forming the information we know today on the origination of our energy source.

The process of photosynthesis consists of two different stages. As sunlight first enters the plants' leaves, the chlorophyll within the chloroplasts absorbs the photons using their green color pigments. After entering the chloroplast, the first stage of photosynthesis known as the light-dependent reaction occurs in the thylakoid membranes, splitting out the electrons within water molecules using light particles that result in the release of oxygen and generating of energy molecules (ATP and NADPH). These energy molecules later on get transported from the thylakoid into the stroma of the chloroplast, where the second stage of photosynthesis takes place called the Calvin Cycle. During the Calvin Cycle, the enzyme Rubisco captures an additional carbon dioxide, forcing it to join groups of five carbon dioxide, creating a six-carbon compound that will immediately get split into two three-carbon compounds called 3-phosphoglycerate (3-PGA). A small portion of the 3-PGA will then exit the cycle and eventually be used as glucose, while others will get recycled back into the stroma to create a positive feed loop of creating more six-carbon compounds to prolong the Calvin Cycle.

Environmental Science

Zichun Wnag

Environmental science is an interdisciplinary field that studies the interactions between environmental systems and human systems. It emerged as a formal discipline in the 1960s, a time when public awareness of environmental issues was minimal. The publication of Rachel Carson's book *Silent Spring* marked a significant turning point, igniting widespread societal concern. In recent years, as advancements in technology and energy development have progressed, the importance of environmental science has become increasingly evident, as the world faces critical challenges such as rising temperatures, ecological destruction, and the overconsumption of resources.

Currently, environmental science primarily revolves around issues like resource usage, particularly non-renewable ones like oil and coal, the pressure of rapid human population growth on the environmental carrying capacity, and pollution. According to the 2022 Global Ocean Environment Report published by the United Nations Environment Programme, the increasing presence of pollution in the oceans, such as plastics, chemical waste, and oil spills, which become severe threats to marine ecosystems. This not only affects the survival of marine species but also endangers human health and livelihoods. These challenges show the urgent need for effective policies and practices. Addressing these issues demands innovative solutions, especially the transition to renewable energy. This shift can significantly reduce our reliance on fossil fuels and we can develop easily accessible natural resources like solar, wind, and hydropower, ultimately lowering greenhouse gas emissions and promoting energy independence. To accelerate the progress of environmental protection, many countries have enacted relevant laws and regulations to impose constraints. Additionally, global agreements like the United Nations Sustainable Development Goals play a crucial role in ensuring the responsible use and management of non-renewable resources.

Cosmic Galaxy

Ethan Liu

The universe is a truly incredible place that inspires curiosity and awe in everyone who takes the time to look up at the night sky. It is estimated that the universe is approximately 13.8 billion years old and is filled with billions of galaxies, each containing countless stars. Among these galaxies, our own Milky Way is home to the Solar System, which includes Earth - the only planet we know of that supports life.

What makes the universe so fascinating is its immensity. When we gaze at the stars, we're looking back in time, because light from distant stars takes millions of years to reach us. That means the light we're seeing tonight could come from stars that exploded long ago! Thanks to powerful telescopes like the Hubble Space Telescope, scientists can explore these distant galaxies and learn more about how they formed and changed over time.

Another mind-blowing aspect of the universe is the concept of dark matter and dark energy. These mysterious forces make up about 95% of the universe, but we can't see them or fully understand what they are. Dark matter does not emit light and is therefore invisible, while dark energy is thought to be responsible for the expansion of the universe.

In short, the Universe is more than just a collection of stars and galaxies; it is a vast, mysterious realm that continues to inspire us. Each new discovery brings us closer to understanding the wonders of the universe and reminds us of our small place in this magnificent universe.

Silent Suffering: Recognizing and Addressing Chronic Migraines

Brandon Pian

A migraine is a headache that can cause severe throbbing pain or a pulsing sensation, usually on one side of the head. Other symptoms of migraines are also nausea, vomiting, and extreme sensitivity to light and sound. The length of migraines varies from hours to days. An aura or a warning symptom occurs before or with the headache. An aura includes visual disturbances such as flashes of light or blind spots and other disturbances such as a tingling on one side of the body or the difficulty of speaking. Different medications such as Nonsteroidal anti-inflammatory drugs, Acetaminophen (Tylenol), and others can help prevent migraines and make them less painful.

Migraines affect people of all ages and progress through four stages, but not everyone goes through all of them.

1. Prodrome

This occurs one or two days before a migraine with symptoms of constipation, mood changes, food cravings, neck stiffness, and more.

2. Aura

The Aura occurs before or during a migraine. Auras are reversible symptoms of the nervous system. Auras include visual phenomena, vision loss, needle sensations in an arm or leg, and more.

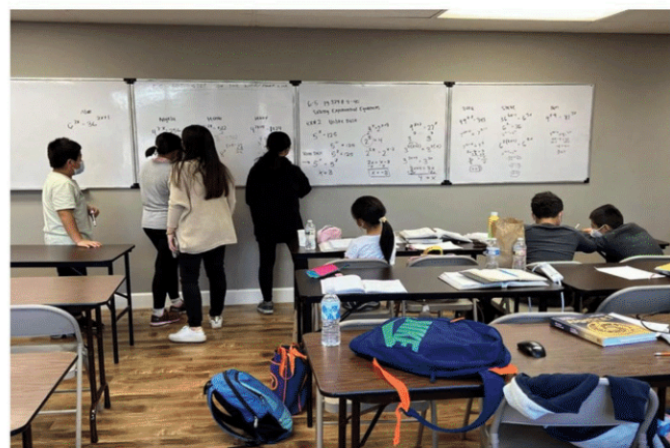
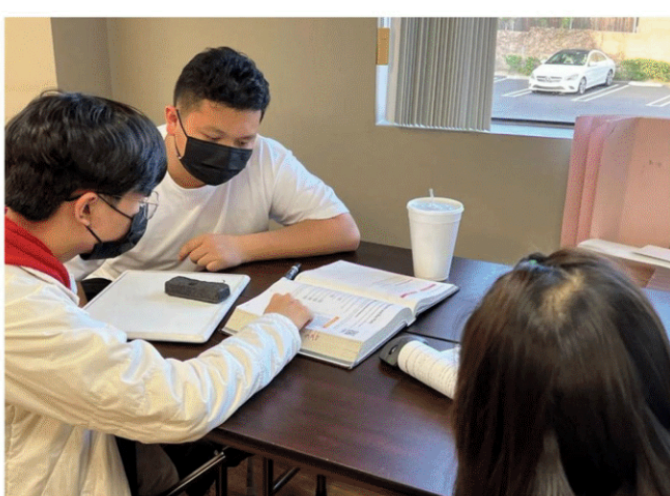
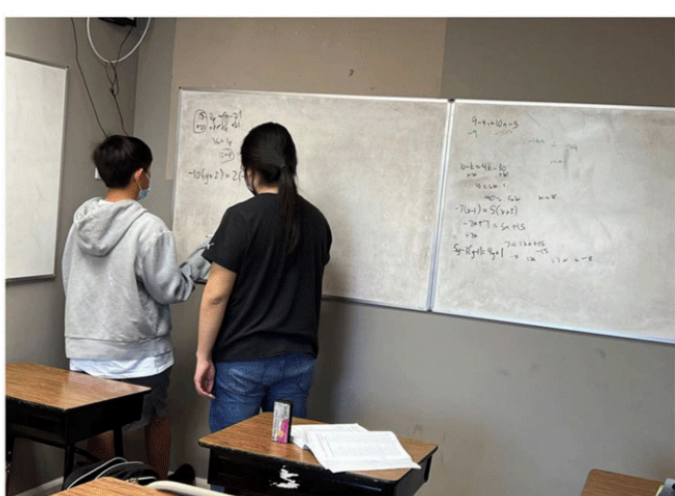
3. Attack

Migraines last from 4 to 72 hours if untreated. During a migraine, people may have pain usually on one side of their head, but often on both sides, pain that throbs or pulses, and more.

4. Post-Drome

After a migraine people feel drained and confused. Sudden head movement might bring pain again briefly.

Though the cause of migraines hasn't been found, researchers found common migraine triggers. Such as emotions, biological and environmental conditions, fatigue, flickering lights, weather changes, or certain foods and drinks.



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AP Physics 1, 2, C
著重公式練習與演算運用, 同時準備SATII應考

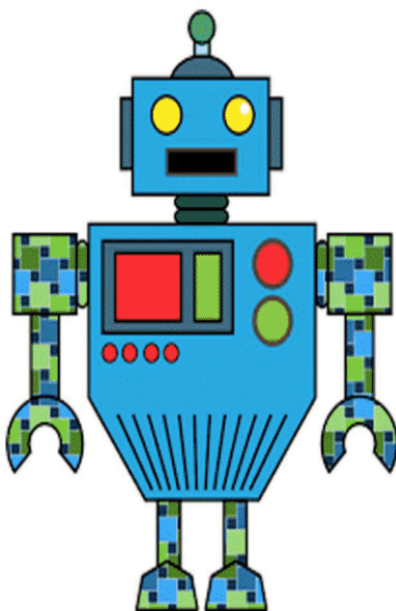
AP Calculus BC, AB
講解清晰海量試題練習 同年應試二科省時省力

數學加強班
Algebra 1, 2 Geometry
物理榮譽班
7-11年級。Honors課程。
為AP物理作充足準備

電腦編程基礎班 Java C++ Python
4-12年級為AP Computer 課程準備
並可參加全國及各項國際AI競賽

SAT 英文寫作班
4-12年級, 閱讀, 文法, 寫作
** 因材施教 突破盲點 **

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